

DM ratio sensitivity — polished draft

BrocaOS (draft)

2025-12-25

Abstract

A more polished draft with figures and methods.

1 Methods

We compute $R = 5 + \mathbb{E}[e^{-\Lambda}]$ under Poisson, Negative Binomial, and Lognormal models for Λ .

2 Results

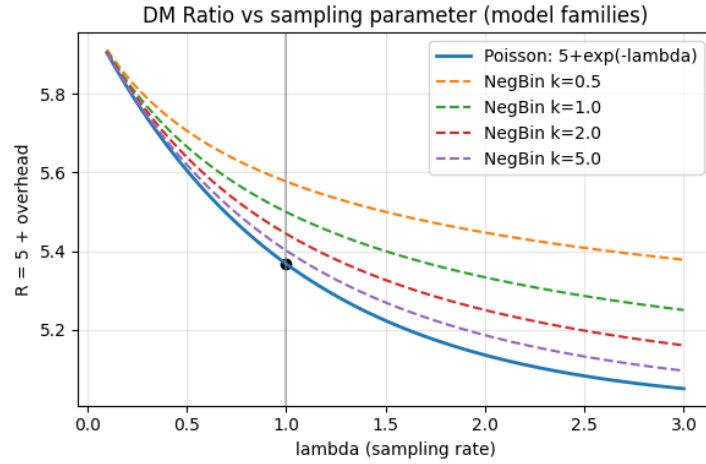


Figure 1: DM ratio as a function of the Poisson mean rate.

3 Numeric summary

Planck observed ratio: 5.364327223961

Poisson (lambda=1): 5.367879441171

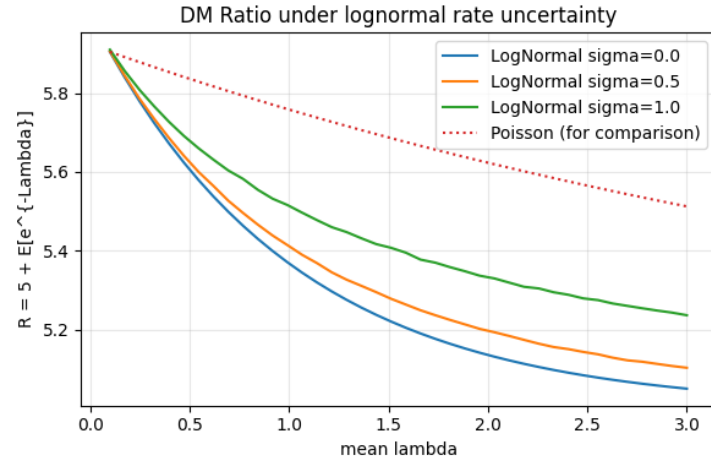


Figure 2: Comparison between Poisson and lognormal-rate uncertainty models.

NegBin k=0.5: 5.577350269190

4 Discussion

Further work listed.